Customer Shopping Behavior Analysis

1. Executive Summary

This project analyzes customer shopping behavior using transactional data from 3,900 purchases. The objective is to identify purchasing patterns, customer segments, product preferences, and subscription trends that can support data-driven business decisions and improve customer engagement strategies.

2. Dataset Overview

Total Records: 3,900
Total Attributes: 18
Key Data Categories:
• Customer demographics (Age, Gender, Location, Subscription Status)
• Purchase details (Product, Category, Amount, Season, Size, Color)
• Behavioral indicators (Discount Usage, Previous Purchases, Review Rating, Shipping Type)
• Missing values identified in the Review Rating attribute.

3. Data Preparation and Processing

The dataset was cleaned and prepared using Python. Key activities included data validation, missing value treatment, column standardization, feature engineering, redundancy removal, and quality verification. Missing review ratings were imputed using category-based median values. Additional analytical features were generated to support customer segmentation and purchasing analysis.

4. Database Integration

After preprocessing, the cleaned dataset was integrated into PostgreSQL using SQLAlchemy and psycopg2. The database environment enabled structured querying, aggregation, customer segmentation, and business intelligence reporting.

5. Business Analysis Performed

• Revenue analysis by customer gender.
• Identification of high-spending customers using discounts.
• Evaluation of top-rated products based on customer reviews.
• Comparison of spending behavior across shipping methods.
• Subscriber versus non-subscriber revenue analysis.
• Identification of products highly dependent on discounts.
• Customer segmentation into New, Returning, and Loyal groups.
• Category-wise top-performing products.
• Relationship analysis between repeat purchases and subscriptions.
• Revenue contribution analysis across age groups.

6. Dashboard Development

Interactive dashboards were developed in Power BI to present key performance indicators, customer insights, revenue trends, and product performance metrics in a visual and business-friendly format.

7. Strategic Recommendations

• Strengthen subscription programs through exclusive benefits and targeted campaigns.
• Implement loyalty initiatives to increase customer retention and repeat purchases.
• Optimize discount strategies to balance profitability and sales growth.
• Promote top-rated and high-performing products through focused marketing efforts.
• Apply age-group-based targeting to improve campaign effectiveness and customer engagement.

8. Conclusion

The project demonstrates how Python, PostgreSQL, and Power BI can be combined to transform raw transactional data into actionable business intelligence. The resulting insights support strategic planning, customer retention, revenue optimization, and data-driven decision making.